

Numerical Problems with Guided Solutions

Diffusion and Non-Equilibrium in Semiconductors

Solid State Physics Lessons

November 7, 2025

Useful constants (300 K unless noted)

- Elementary charge: $q = 1.602 \times 10^{-19} \text{ C}$
- Thermal voltage: $kT/q \approx 0.0259 \text{ V}$ at 300 K
- Intrinsic carrier concentration: $n_i(\text{Si}, 300 \text{ K}) \approx 1.0 \times 10^{10} \text{ cm}^{-3}$

1 Diffusion in Semiconductors (5 problems)

1. Mixed drift–diffusion currents (Einstein relation)

Problem. In uniformly doped Si at 300 K, take $\mu_n = 1400 \text{ cm}^2/(\text{V s})$, $\mu_p = 450 \text{ cm}^2/(\text{V s})$. Let the electric field be $E = -80 \text{ V/cm}$ (negative means pointing in $-x$), carrier concentrations $n = 8.0 \times 10^{15} \text{ cm}^{-3}$, $p = 2.0 \times 10^{14} \text{ cm}^{-3}$ with gradients $\frac{dn}{dx} = -6.0 \times 10^{19} \text{ cm}^{-4}$ and $\frac{dp}{dx} = +2.5 \times 10^{19} \text{ cm}^{-4}$ along $+x$. Using $D = \mu(kT/q)$, compute J_n and J_p (A/cm^2). State the drift and diffusion parts separately.

Solution.

Step 1: Calculate diffusion coefficients using the Einstein relation.

The Einstein relation connects diffusivity to mobility:

$$D_n = \mu_n \frac{kT}{q}, \quad D_p = \mu_p \frac{kT}{q}$$

At 300 K, $kT/q = 0.0259 \text{ V}$. Therefore:

$$D_n = 1400 \times 0.0259 = 36.26 \text{ cm}^2/\text{s}, \quad D_p = 450 \times 0.0259 = 11.655 \text{ cm}^2/\text{s}.$$

Step 2: Calculate drift currents.

The drift current arises from the electric field acting on mobile carriers:

$$J_{n,\text{drift}} = qn\mu_n E, \quad J_{p,\text{drift}} = qp\mu_p E.$$

For electrons:

$$J_{n,\text{drift}} = (1.602 \times 10^{-19}) \times (8.0 \times 10^{15}) \times (1400) \times (-80) = \boxed{-143.5 \text{ A/cm}^2}.$$

For holes:

$$J_{p,\text{drift}} = (1.602 \times 10^{-19}) \times (2.0 \times 10^{14}) \times (450) \times (-80) = \boxed{-1.15 \text{ A/cm}^2}.$$

The negative sign indicates currents flow in the $-x$ direction (opposite to the applied field direction). The electron drift current is much larger because $n \gg p$.

Step 3: Calculate diffusion currents.

Diffusion currents arise when carriers flow down concentration gradients:

$$J_{n,\text{diff}} = qD_n \frac{dn}{dx}, \quad J_{p,\text{diff}} = -qD_p \frac{dp}{dx}.$$

(Note the minus sign for holes: they diffuse opposite to ∇p , following the direction of ∇n .)

For electrons:

$$J_{n,\text{diff}} = (1.602 \times 10^{-19}) \times (36.26) \times (-6.0 \times 10^{19}) = \boxed{-348.5 \text{ A/cm}^2}.$$

For holes:

$$J_{p,\text{diff}} = -(1.602 \times 10^{-19}) \times (11.655) \times (2.5 \times 10^{19}) = \boxed{-46.7 \text{ A/cm}^2}.$$

Step 4: Calculate total currents.

$$J_n = J_{n,\text{drift}} + J_{n,\text{diff}} = (-143.5) + (-348.5) = \boxed{-492.0 \text{ A/cm}^2}.$$

$$J_p = J_{p,\text{drift}} + J_{p,\text{diff}} = (-1.15) + (-46.7) = \boxed{-47.8 \text{ A/cm}^2}.$$

Summary: Both electrons and holes carry current in the $-x$ direction. For electrons, diffusion dominates; for holes, drift is negligible. The drift-diffusion equation successfully combines both transport mechanisms.

2. Built-in field in a graded n -type bar (zero-current condition)

Problem. An n -type Si bar at 300 K has $n(x) = N_D(x) = N_0 e^{ax}$ with $N_0 = 1 \times 10^{16} \text{ cm}^{-3}$ and $a = 1.0 \times 10^3 \text{ cm}^{-1}$. Assuming low-level majority transport and $\mu_n = 1350 \text{ cm}^2/(\text{V s})$, find the electric field $E(x)$ needed so that $J_n = 0$ everywhere.

Solution.

Step 1: Apply the zero-current condition.

Setting $J_n = 0$ in the drift-diffusion equation:

$$0 = q(n\mu_n E + D_n \frac{dn}{dx}).$$

Step 2: Solve for the electric field.

Rearranging:

$$E = -\frac{D_n}{\mu_n} \frac{1}{n} \frac{dn}{dx}.$$

Applying the Einstein relation, $D_n/\mu_n = kT/q$:

$$E(x) = -\frac{kT}{q} \frac{1}{n(x)} \frac{dn}{dx}.$$

This field exactly balances drift and diffusion—it is the *equilibrium built-in field*.

Step 3: Calculate the spatial derivative.

Given $n(x) = N_0 e^{ax}$, the logarithmic derivative is:

$$\frac{1}{n} \frac{dn}{dx} = \frac{d}{dx} \ln n(x) = \frac{d}{dx}(ax) = a.$$

Step 4: Substitute and find the field.

$$E(x) = -\frac{kT}{q} \cdot a = -0.0259 \times (1.0 \times 10^3) = \boxed{-25.9 \text{ V/cm}}.$$

Key result: The built-in field is *uniform* (independent of x), even though the doping varies exponentially! This uniform field arises because it must be proportional to $\ln(n)$. The negative sign indicates the field opposes the direction of increasing doping.

3. Pure diffusion current magnitudes

Problem. At 300 K, take $D_n = 35 \text{ cm}^2/\text{s}$, $D_p = 12 \text{ cm}^2/\text{s}$. If $\frac{dn}{dx} = +1.2 \times 10^{20} \text{ cm}^{-4}$ and $\frac{dp}{dx} = -8.0 \times 10^{19} \text{ cm}^{-4}$, compute $J_{n,\text{diff}}$ and $J_{p,\text{diff}}$ (A/cm^2) and their directions.

Solution.

Step 1: Compute electron diffusion current.

Electrons diffuse down their concentration gradient according to:

$$J_{n,\text{diff}} = qD_n \frac{dn}{dx}.$$

Since $dn/dx > 0$, electron concentration increases in the $+x$ direction. Electrons diffuse from high to low concentration, so they flow in the $+x$ direction.

$$J_{n,\text{diff}} = (1.602 \times 10^{-19}) \times (35) \times (1.2 \times 10^{20}) = \boxed{673 \text{ A/cm}^2} \quad (\text{along } +x).$$

Step 2: Compute hole diffusion current.

Holes diffuse opposite to the hole concentration gradient:

$$J_{p,\text{diff}} = -qD_p \frac{dp}{dx}.$$

Since $dp/dx < 0$, hole concentration decreases in the $+x$ direction. Holes diffuse from high to low concentration (toward $+x$).

$$J_{p,\text{diff}} = -(1.602 \times 10^{-19}) \times (12) \times (-8.0 \times 10^{19}) = \boxed{154 \text{ A/cm}^2} \quad (\text{along } +x).$$

Physical insight: Both carriers flow in the same direction ($+x$), but for opposite reasons. This cooperative transport is characteristic of the *ambipolar diffusion* regime, where the two carrier types move together under the influence of a built-in electric field created by their concentration imbalance.

4. Ambipolar diffusion coefficient and length

Problem. In moderately doped n -Si at 300 K, $n_0 = 5.0 \times 10^{15} \text{ cm}^{-3}$, $p_0 = n_i^2/n_0$ with $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Mobilities: $\mu_n = 1200 \text{ cm}^2/(\text{V s})$, $\mu_p = 450 \text{ cm}^2/(\text{V s})$. Minority lifetime $\tau_p = 2.0 \mu\text{s}$, majority lifetime $\tau_n = 30 \mu\text{s}$. Compute (i) D_n, D_p ; (ii) ambipolar diffusion coefficient D_a defined by

$$D_a = \frac{D_n \mu_p p_0 + D_p \mu_n n_0}{\mu_n n_0 + \mu_p p_0},$$

(iii) effective ambipolar lifetime

$$\tau_a = \frac{n_0 + p_0}{n_0/\tau_n + p_0/\tau_p},$$

and (iv) ambipolar diffusion length $L_a = \sqrt{D_a \tau_a}$.

Solution.

Step 1: Calculate diffusion coefficients using Einstein relation.

$$D_n = \mu_n \frac{kT}{q} = 1200 \times 0.0259 = 31.08 \text{ cm}^2/\text{s}.$$

$$D_p = \mu_p \frac{kT}{q} = 450 \times 0.0259 = 11.65 \text{ cm}^2/\text{s}.$$

Step 2: Determine equilibrium carrier concentrations.

Majority carriers (electrons): $n_0 = 5.0 \times 10^{15} \text{ cm}^{-3}$ (given).

Minority carriers (holes):

$$p_0 = \frac{n_i^2}{n_0} = \frac{(1.0 \times 10^{10})^2}{5.0 \times 10^{15}} = 2.0 \times 10^4 \text{ cm}^{-3}.$$

Note that $p_0 \ll n_0$, characteristic of doped material.

Step 3: Calculate ambipolar diffusion coefficient.

The ambipolar diffusivity represents a weighted average of individual diffusivities, accounting for the coupled motion of both carrier types. Compute the denominator (related to total conductivity):

$$\mu_n n_0 + \mu_p p_0 = (1200)(5.0 \times 10^{15}) + (450)(2.0 \times 10^4) = 6.0 \times 10^{18} + 9.0 \times 10^6 \approx 6.0 \times 10^{18}.$$

The second term is negligible.

Compute the numerator (weighted diffusivity sum):

$$\begin{aligned} D_n \mu_p p_0 + D_p \mu_n n_0 &= (31.08)(450)(2.0 \times 10^4) + (11.65)(1200)(5.0 \times 10^{15}) \\ &= 2.80 \times 10^8 + 6.99 \times 10^{19} \approx 7.0 \times 10^{19}. \end{aligned}$$

Therefore:

$$D_a = \frac{7.0 \times 10^{19}}{6.0 \times 10^{18}} = 11.67 \text{ cm}^2/\text{s}.$$

Important: $D_a \approx D_p$. In n -type material, the ambipolar diffusion is limited by the minority carrier (hole) properties. This is because holes are less mobile and present in lower concentration, creating the bottleneck for joint diffusion.

Step 4: Calculate effective ambipolar lifetime.

The ambipolar lifetime is the harmonic mean of the bulk and minority lifetimes, weighted by carrier concentrations:

$$\tau_a = \frac{n_0 + p_0}{\frac{n_0}{\tau_n} + \frac{p_0}{\tau_p}}.$$

Numerator: $n_0 + p_0 = 5.0 \times 10^{15} + 2.0 \times 10^4 \approx 5.0 \times 10^{15}$ (first term dominates).

Denominator:

$$\begin{aligned} \frac{n_0}{\tau_n} + \frac{p_0}{\tau_p} &= \frac{5.0 \times 10^{15}}{30 \times 10^{-6}} + \frac{2.0 \times 10^4}{2.0 \times 10^{-6}}. \\ &= 1.67 \times 10^{20} + 1.0 \times 10^{10} \approx 1.67 \times 10^{20} \quad (\text{first term dominates}). \end{aligned}$$

Therefore:

$$\tau_a = \frac{5.0 \times 10^{15}}{1.67 \times 10^{20}} = 3.0 \times 10^{-5} \text{ s} = \boxed{30 \mu\text{s}}.$$

Observation: $\tau_a \approx \tau_n$ because the majority carriers (electrons) control the lifetime in this material.

Step 5: Calculate ambipolar diffusion length.

The diffusion length characterizes how far carriers can travel by diffusion before recombining:

$$L_a = \sqrt{D_a \tau_a} = \sqrt{(11.67) \times (30 \times 10^{-6})} = \sqrt{3.50 \times 10^{-4}} = \boxed{0.0187 \text{ cm} = 187 \mu\text{m}}.$$

This length scale is important for device design: regions smaller than L_a allow efficient collection of diffusing carriers.

5. Fundamental diffusion time across a slab

Problem. Excess carriers in a slab of thickness $W = 200 \mu\text{m}$ (one-dimensional, between two ideal sinks) diffuse with $D = 25 \text{ cm}^2/\text{s}$. Estimate the dominant decay time constant τ_1 (first mode) due to diffusion alone:

$$\tau_1 \approx \frac{W^2}{\pi^2 D}.$$

Solution.

Step 1: Understand the physics.

When excess carriers are introduced uniformly across a slab bounded by ideal recombination sinks, they redistribute and decay according to the diffusion equation. The slowest-decaying mode (the one that determines the overall lifetime) is the fundamental mode with spatial profile $\sin(\pi x/W)$. The decay time is given by the relation above.

Step 2: Convert units.

Convert thickness to centimeters:

$$W = 200 \mu\text{m} = 200 \times 10^{-4} \text{ cm} = 2.0 \times 10^{-2} \text{ cm}.$$

Step 3: Calculate the decay time.

$$\tau_1 = \frac{(2.0 \times 10^{-2})^2}{\pi^2 \times 25} = \frac{4.0 \times 10^{-4}}{246.74} = \boxed{1.62 \times 10^{-6} \text{ s} = 1.62 \mu\text{s}}.$$

Physical interpretation: Excess carriers decay within microseconds due to diffusion-limited recombination. For thicker slabs, $\tau_1 \propto W^2$, showing that diffusion time scales with the square of the distance.

2 Non-Equilibrium in Semiconductors (5 problems)

6. Steady-state excess and photoconductivity

Problem. Uniform generation $g = 6.0 \times 10^{19} \text{ cm}^{-3} \text{ s}^{-1}$ is sustained in Si (300 K). Under low-level injection with $\tau_n = \tau_p = \tau = 0.8 \mu\text{s}$, $\mu_n = 1300 \text{ cm}^2/(\text{V s})$, $\mu_p = 500 \text{ cm}^2/(\text{V s})$, find (i) $\Delta n = \Delta p$ and (ii) $\Delta\sigma = q(\Delta n\mu_n + \Delta p\mu_p)$.

Solution.

Step 1: Find steady-state excess carrier concentration.

Under steady-state illumination, the generation rate is balanced by recombination:

$$g = \frac{\Delta n}{\tau} \Rightarrow \Delta n = g\tau.$$

By charge neutrality in the bulk, $\Delta n = \Delta p$.

$$\Delta n = \Delta p = (6.0 \times 10^{19})(0.8 \times 10^{-6}) = \boxed{4.8 \times 10^{13} \text{ cm}^{-3}}.$$

Step 2: Calculate the photoconductivity.

Excess carriers increase the conductivity:

$$\Delta\sigma = q\Delta n(\mu_n + \mu_p) = q\Delta n(\mu_n + \mu_p).$$

The sum of mobilities accounts for both carrier types contributing to current.

$$\begin{aligned} \Delta\sigma &= (1.602 \times 10^{-19}) \times (4.8 \times 10^{13}) \times (1300 + 500). \\ &= (1.602 \times 10^{-19}) \times (4.8 \times 10^{13}) \times (1800) = \boxed{1.38 \times 10^{-2} \Omega^{-1} \text{ cm}^{-1}}. \end{aligned}$$

Summary: The photocurrent creates both electrons and holes at equal rates. Both carrier types contribute to the increased conductivity. This is the basis for photodetectors and solar cells.

7. Turn-off transient of excess carriers

Problem. At $t = 0$, light is switched off in p -type Si, leaving a uniform initial excess $\Delta n(0) = 2.0 \times 10^{13} \text{ cm}^{-3}$. If $\tau = 3.0 \mu\text{s}$, find (i) $\Delta n(t)$, (ii) time for Δn to drop to 10% of its initial value, and (iii) the time-dependent photoconductivity $\Delta\sigma(t)$ given $\mu_n = 1100 \text{ cm}^2/(\text{V s})$, $\mu_p = 400 \text{ cm}^2/(\text{V s})$.

Solution.

Step 1: Write the recombination kinetics.

Without generation, excess carriers decay through recombination at a first-order rate:

$$\frac{d(\Delta n)}{dt} = -\frac{\Delta n}{\tau}.$$

This differential equation has the solution:

$$\Delta n(t) = \Delta n(0)e^{-t/\tau}.$$

Step 2: Find the time for decay to 10%.

Set $\Delta n(t) = 0.1 \times \Delta n(0)$:

$$0.1 = e^{-t/\tau} \Rightarrow t = \tau \ln(10) = (3.0 \mu\text{s}) \times 2.3026 = \boxed{6.91 \mu\text{s}}.$$

Step 3: Express time-dependent photoconductivity.

The excess conductivity evolves as:

$$\Delta\sigma(t) = q\Delta n(t)(\mu_n + \mu_p) = q\Delta n(0)e^{-t/\tau}(\mu_n + \mu_p).$$

Initial value at $t = 0$:

$$\Delta\sigma(0) = (1.602 \times 10^{-19}) \times (2.0 \times 10^{13}) \times (1100 + 400) = 4.81 \times 10^{-3} \Omega^{-1}\text{cm}^{-1}.$$

Therefore:

$$\Delta\sigma(t) = 4.81 \times 10^{-3} \Omega^{-1}\text{cm}^{-1} \times \exp\left(-\frac{t}{3.0 \mu\text{s}}\right).$$

Practical note: This exponential decay is observed in transient photoconductivity measurements, which are used to extract minority carrier lifetimes from experimental data.

8. Quasi-Fermi level separation from excess carriers

Problem. An n -type Si sample at 300 K has $n_0 = 1.0 \times 10^{16} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Under low-level injection, uniform excess $\Delta n = \Delta p = 5.0 \times 10^{13} \text{ cm}^{-3}$ is present. Compute the product np , the ratio np/n_i^2 , and the quasi-Fermi level separation $\Delta E_F = E_{Fn} - E_{Fp} = kT \ln\left(\frac{np}{n_i^2}\right)$ in eV.

Solution.

Step 1: Determine total carrier concentrations.

Electrons: The initial electron concentration is $n_0 = 1.0 \times 10^{16} \text{ cm}^{-3}$ (given). Adding excess:

$$n = n_0 + \Delta n = 1.0 \times 10^{16} + 5.0 \times 10^{13} \approx 1.005 \times 10^{16} \text{ cm}^{-3}.$$

Holes: The initial hole concentration is (from $n_0 p_0 = n_i^2$):

$$p_0 = \frac{n_i^2}{n_0} = \frac{(1.0 \times 10^{10})^2}{1.0 \times 10^{16}} = 1.0 \times 10^4 \text{ cm}^{-3}.$$

Adding excess:

$$p = p_0 + \Delta p = 1.0 \times 10^4 + 5.0 \times 10^{13} \approx 5.0 \times 10^{13} \text{ cm}^{-3}.$$

(The excess carriers dominate the hole concentration.)

Step 2: Calculate the product np .

$$np = (1.005 \times 10^{16}) \times (5.0 \times 10^{13}) = 5.025 \times 10^{29} \text{ cm}^{-6}.$$

Step 3: Compute the mass action product ratio.

Under non-equilibrium, the mass action law becomes:

$$np = n_i^2 \exp\left(\frac{\Delta E_F}{kT}\right),$$

where $\Delta E_F = E_{Fn} - E_{Fp}$ is the separation of quasi-Fermi levels.

$$\frac{np}{n_i^2} = \frac{5.025 \times 10^{29}}{(1.0 \times 10^{10})^2} = \frac{5.025 \times 10^{29}}{1.0 \times 10^{20}} = \boxed{5.03 \times 10^9}.$$

Step 4: Calculate quasi-Fermi level separation.

$$\Delta E_F = kT \ln \left(\frac{np}{n_i^2} \right) = (0.0259) \ln(5.03 \times 10^9).$$

Compute the logarithm:

$$\ln(5.03 \times 10^9) = \ln(5.03) + 9 \ln(10) = 1.615 + 9(2.303) = 1.615 + 20.727 = 22.342.$$

Therefore:

$$\Delta E_F = 0.0259 \times 22.342 = \boxed{0.579 \text{ eV}}.$$

Physical meaning: The quasi-Fermi level separation reflects how far the material is driven from equilibrium. At equilibrium, $\Delta E_F = 0$. Here, the 0.579 eV separation indicates substantial non-equilibrium. This splitting is exploited in photodiodes and solar cells to generate output voltage.

9. Effective lifetime with surface recombination (approximate)

Problem. A Si slab of thickness $W = 300 \mu\text{m}$ has bulk lifetime $\tau_b = 20 \mu\text{s}$ and ambipolar diffusion coefficient $D = 15 \text{ cm}^2/\text{s}$. Both surfaces are high-recombination sinks. Estimate the effective lifetime under uniform generation using

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_b} + \frac{\pi^2 D}{W^2} \quad (\text{first-mode approximation}).$$

Compute τ_{eff} .

Solution.

Step 1: Understand the two recombination mechanisms.

The effective lifetime combines two independent processes:

- Bulk recombination: characterized by τ_b , uniform throughout the volume.
- Surface recombination: localized at boundaries; carriers must diffuse to surfaces to recombine.

The reciprocal lifetimes add: $1/\tau_{\text{eff}} = 1/\tau_b + 1/\tau_{\text{surface}}$.

Step 2: Convert units.

$$W = 300 \mu\text{m} = 300 \times 10^{-4} \text{ cm} = 3.0 \times 10^{-2} \text{ cm}.$$

Step 3: Calculate bulk recombination rate.

$$\frac{1}{\tau_b} = \frac{1}{20 \times 10^{-6}} = 5.0 \times 10^4 \text{ s}^{-1}.$$

Step 4: Calculate surface recombination rate (diffusion contribution).

The rate at which carriers reach the surface by diffusion is proportional to $\pi^2 D/W^2$ (from the fundamental mode of the diffusion equation):

$$\frac{\pi^2 D}{W^2} = \frac{\pi^2 \times 15}{(3.0 \times 10^{-2})^2} = \frac{9.8696 \times 15}{9.0 \times 10^{-4}} = \frac{148.044}{9.0 \times 10^{-4}} = 1.644 \times 10^5 \text{ s}^{-1}.$$

Step 5: Calculate effective lifetime.

$$\frac{1}{\tau_{\text{eff}}} = 5.0 \times 10^4 + 1.644 \times 10^5 = 2.144 \times 10^5 \text{ s}^{-1}.$$

$$\tau_{\text{eff}} = \frac{1}{2.144 \times 10^5} = \boxed{4.66 \mu\text{s}}.$$

Observation: Surface recombination dominates. The effective lifetime is much shorter than the bulk lifetime ($4.66 \mu\text{s}$ vs $20 \mu\text{s}$) because carriers can easily reach the high-recombination surfaces. In practice, surface passivation (creating a native oxide layer or other protective coating) is crucial to reduce surface recombination and extend device lifetime.

10. Dielectric relaxation time vs recombination

Problem. For illuminated p -type Si at 300 K, majority hole concentration $p_0 = 5.0 \times 10^{15} \text{ cm}^{-3}$, mobility $\mu_p = 480 \text{ cm}^2/(\text{V s})$, permittivity $\varepsilon = \varepsilon_s \approx 11.7\varepsilon_0$ with $\varepsilon_0 = 8.85 \times 10^{-14} \text{ F/cm}$.

(i) Compute conductivity $\sigma \approx q\mu_p p_0$ and dielectric relaxation time $\tau_d = \varepsilon/\sigma$. (ii) Compare τ_d to a typical recombination lifetime $\tau = 5 \mu\text{s}$ to justify quasi-neutrality.

Solution.

Step 1: Calculate the conductivity.

For a p -type material dominated by hole transport:

$$\sigma = q\mu_p p_0 = (1.602 \times 10^{-19}) \times (480) \times (5.0 \times 10^{15}).$$

Compute step-by-step:

$$480 \times 5.0 = 2400, \quad 2400 \times 10^{15} = 2.4 \times 10^{18}.$$

$$\sigma = (1.602 \times 10^{-19}) \times (2.4 \times 10^{18}) = \boxed{0.384 \Omega^{-1} \text{ cm}^{-1}}.$$

Step 2: Calculate the permittivity.

The static permittivity of Si is:

$$\varepsilon = 11.7 \times \varepsilon_0 = 11.7 \times (8.85 \times 10^{-14}) = \boxed{1.035 \times 10^{-12} \text{ F/cm}}.$$

Step 3: Calculate dielectric relaxation time.

The dielectric relaxation time characterizes the timescale on which charge imbalances are neutralized by current flow. It is defined as:

$$\tau_d = \frac{\varepsilon}{\sigma} = \frac{1.035 \times 10^{-12}}{0.384} = \boxed{2.69 \times 10^{-12} \text{ s}} = \boxed{2.7 \text{ ps}}.$$

Step 4: Compare time scales.

- Dielectric relaxation time: $\tau_d \approx 2.7 \text{ ps}$ (picoseconds).
- Recombination lifetime: $\tau \approx 5 \mu\text{s}$ (microseconds).
- Ratio: $\tau_d/\tau \approx 10^{-6}$.

Physical interpretation: The dielectric relaxation is vastly faster than recombination. This means any local charge imbalance is neutralized almost instantaneously compared to the timescale of carrier generation and recombination. Therefore, the *quasi-neutral approximation*—assuming $n \approx p + N_D$ locally throughout the material—is justified. Charge neutrality is maintained continuously as carriers are generated and recombine, making the quasi-neutral transport equations valid.

Key insight: Semiconductor transport can be divided into two regimes:

1. Fast (τ_d): Electrostatic equilibration via current flow.
2. Slow (τ): Carrier dynamics via generation/recombination and diffusion.

This separation of timescales simplifies the analysis of semiconductor devices and justifies the quasi-neutral approximation used throughout semiconductor physics.

Practical implications:

- In device operation, we can assume quasi-neutrality even during transient events (switching, pulsed operation).
- The quasi-neutral equations ($J_n + J_p = J_{\text{total}}$ constant; $n - p = N_D - N_A$ locally) are valid as long as we're concerned with timescales $\gg \tau_d$.
- Device performance is limited by the slow timescale τ , not the electrostatic timescale τ_d .